

DutchSemCor



NWO Geesteswetenschappen
Investeringssubsidie
Middel/Groot
2009-2012

Program

13.00 – 13.20 Welcome (“Galerij”)

13.30 – 15.00 Methodology and tools (“Stoa”)

- Opening by Piek Vossen
- Methodology, approaches and tools (Attila Görög)
- Manual annotation user experiences (Marlisa Hommel & Chantal van Son)
- Semantic Annotation Tool (Fons Laan)
- Manual annotation statistics (Attila Görög)

15.00 – 15.30 Coffee break (“Galerij”)

15.30 – 17.00 WSD systems and results (“Stoa”)

- Corpora and the FoLiA format (Maarten van Gompel)
- Named Entity Recognition using DutchSemCor data (Fons Laan & Edgar Meij)
- WSD, Domain classification and results (R. Izquierdo)
- Closing by Antal van den Bosch

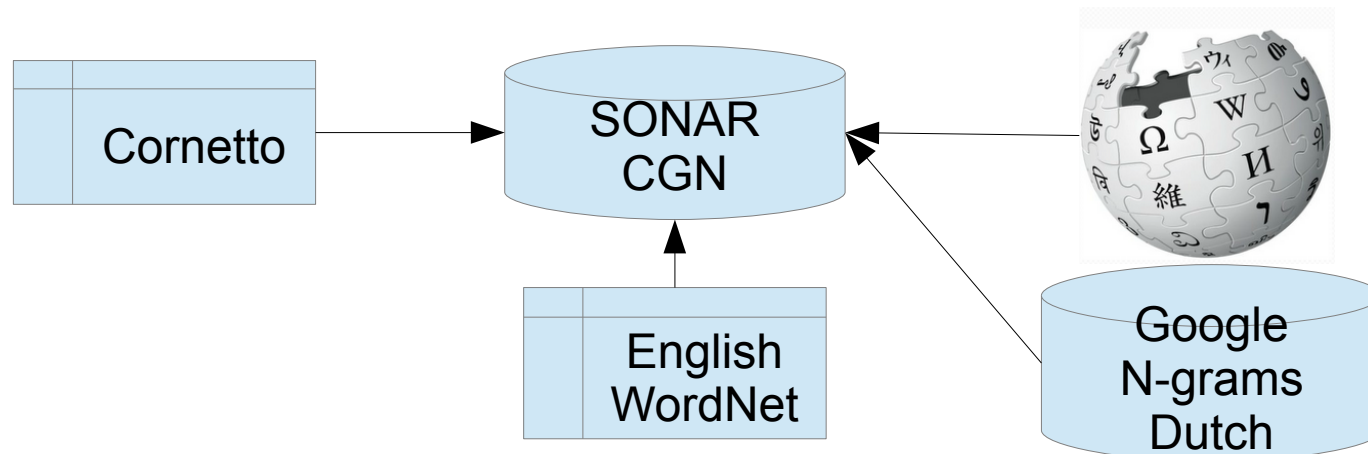
17.15 – 18.00 Drinks (“Galerij”)

Goals

- Deliver a Dutch corpus enriched with semantic tags:
 - Senses of the most-frequent and most-polysemous words;
 - Domain classification;
 - Named entities and Wikipedia links;
- 1 million sense-tagged tokens:
 - 250K tagged by two annotators
 - 750K tagged by one-annotator and machines through active learning

Semantic corpus to enable research

- Study the relation between form (language structure) and meaning;
- Idioms, figurative usage, new meanings, abstract/concrete, metaphors
- Pair distributional semantics with lexical semantics;
- Study relation between text genres and domains and meaning;
- Further improve Word-Sense-Disambiguation
- Exploit linked open data:



Example

Kasparov koos een andere methode om het paard op f5 lastig te vallen.

<http://www.volkskrant.nl/vk/nl/2844/Archief/archief/article/detail/733335/2003/10/11/Soltis-s>

Domains:

<domain id="biol" value="-0.719875"/>
<domain id="media" value="-0.851422"/>
<domain id="pol" value="-0.892249"/>
<domain id="kunst" value="-0.948188"/>
<domain id="mil" value="-0.975102"/>
<domain id="taal" value="-0.976206"/>
<domain id="trans" value="-0.97691"/>
<domain id="handel" value="-0.979618"/>
<domain id="sp" value="-0.985876"/>
<domain id="landb" value="-0.999767"/>

WR-P-P-G-0000148955.p.28.s.3.w.1 Kasparov
Entity

Phrase: Kasparov

Wiki_title: Garri_kasparov

Score: 0.8

Wiki_version: nlwiki-20111104

WR-P-P-G-0000148955.p.28.s.3.w.8 paard
Senses

Timbl: r_n-27277#1.00 r_n27278#0.00

SVM: r_n-27278#0.62 r_n27277#0.06

UKB: r_n-27278#0.83 r_n27277#0.06

Overall results

	Lem-mas	Poly-semy	Anno-tated sense	Double manual annota-tions	Automatic annota-tions	Acc. Folded cross	Acc. random	Acc. all-words
Nouns	1.943	4	94,20%	222.178	18.519.944	88,65	66,92 (71,42)	66,09 (71,81)
Verbs	684	5	97,66%	117.625	23.960.557	87,60	60,55 (63,41)	45,68 (66,43)
Adjectives	360	3	95,00%	33.042	5.317.183	83,06	55,11 (56,88)	52,24 (69,43)
TOTAL	2.987			372.845	47.797.684			

Monosemous senses: 60,833 nouns, 5,593 verbs, 981 adjectives.
Sense-groups: 2,808 groups involving 6,903 lexical units
→ 11276 lexical units reduced to
7181 concepts (36% reduction)

More results

- Cornetto 2.0
- 3 WSD systems
- WSD evaluation data
- Named entity recognizers and wikification
- SAT tool
- Etc...

www2.let.vu.nl/oz/cltl/dutchsemcor/results.html

ANNOTATED DATA

- [HUMAN](#)
- [HUMAN-FILTERED](#)
- [ALL-WORDS](#)
- [MACHINE-SONAR](#)
- [HUMAN-SONAR-RANDOM-EVALUATION](#)
- [MACHINE-DOMAINS](#)
- [MACHINE-IDIOMS-DSC-XML](#)
- [MACHINE-NAMED-ENTITIES](#)

SOFTWARE AND TOOLS

- [SAT](#)
- [CORPUS-INDEXING](#)
- [TIMBL-WSD](#)
- [SVM-BC-WSD](#)
- [UKB-WSD](#)
- [DOMAIN-CLASSIFIER](#)
- [IDIOM TAGGER](#)
- [NE-CLASSIFIER AND WIKIFICATION](#)
- [LOGANALYSER](#)
- [CORNETTO CONVERSION](#)

EVALUATION RESULTS

- [DOMAINS](#)
- [WSD \(TIMBL, SVM-BC, UKB AND VOTING\)](#)
- [NER](#)

DERIVED DATA

- [TARGET-WORDS MATRIX](#)
- [HUMAN ANNOTATION STATISTICS](#)
- [CORPUS STATISTICS](#)
- [UKB RELATIONS DERIVED FROM ANNOTATIONS](#)
- [SENSE-GROUPS](#)
- [BASE CONCEPTS](#)

LEXICAL RESOURCES

- [CORNETTO-2.0](#)
- [UKB RESOURCES](#)

Team-work

- Antal van den Bosch (Univ.Tilburg/Nijmegen)
- Attila Görög (VU)
- Maarten van Gompel (Univ.Tilburg/Nijmegen)
- Rubén Izquierdo Beviá (Univ.Tilburg)
- Fons Laan (UVA)
- Edgar Meij (UVA)
- Maarten de Rijke (UVA)
- Piek Vossen (VU)
- Wouter Weerkamp (UVA)

VU Student assistants

Anneleen Schoen

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Gratia Bruining

Jonica Kaagman

Laura Kipp

Lisanne Ranzijn

Marlisa Hommel

Wilma van Velzen

Charlotte van Tongeren

Daphne van Kessel

Chantal van Son

Sam Vossen

Niqee Vossen

Rosa Scheffer

Milou Kerkhof